

IT367 – SOFTWARE PROJECT MANAGEMENT

LAB 5: EFFORT, DURATION AND COST ESTIMATION

Project Title	Abhinay
Group No. / Division	Group 1
Student Names & Roll Nos.	Ishan - (202452321) - Project Manager Happy - (202452319) - Assistant Project Manager Priyadarshini - (202452324) - scope & requirement manager Kshithesh - (202452329) - configuration and change manager nikitha - (202452336) - Risk manager Triya - (202452338) -> resource & communication manager Arjun - (202452308) - Client Abhishek - (202452339) - Schedule manager Yuvaraju - (202452325) - Estimation & Cost Manager Himanshu - (202452320) - Quality Manager

Objective

To estimate the effort, development duration, average staffing and approximate cost of your own software project using Basic COCOMO.

Step 1 – Software Size from Lab 4

Total estimated size = 4750 LOC

$KLOC = LOC \div 1000 = 4.75 \text{ KLOC}$

Step 2 – Select COCOMO Mode

Selected mode: ☒ Organic ☐ Semi-detached ☐ Embedded

Reason for selecting this mode (1–2 lines):

Abhinay is a relatively small (4.75 KLOC) project with a well-understood, familiar web-application domain (casting/profile management), a small team, and no unusual technical or interface constraints.

This matches the Organic mode profile, so it was selected.

Step 3 – Write the Coefficients

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38

Step 4 – Effort Estimation

Formula: $E = a \times (KLOC)^b$

Substitution: $E = 2.4 \times (4.75)^{1.05}$

Estimated Effort = 12.32 person-months

Step 5 – Development Duration

Formula: $D = c \times (E)^d$

Substitution: $D = 2.5 \times (12.32)^{0.38}$

Estimated Duration = 6.49 months

Step 6 – Average Staffing

Formula: Average Staffing = $E \div D$

Calculation: $12.32 \div 6.49 \approx 1.90$

Average Staffing = 1.90 persons

Step 7 – Approximate Project Cost

Cost per person-month = ₹ 50,000 (assumed)

Formula: Project Cost = Effort $\times$ Cost per person-month

Calculation: $12.32 \times ₹50,000 = ₹6,16,000$

Estimated Project Cost = ₹ 6,16,000 (approx.)

Step 8 – Interpretation

Does the estimate look realistic for your project? ☐ Yes ☐ No ☒ Partly

The estimated effort of 12.32 person-months and development duration of approximately 6.49 months provide a reasonable baseline for planning the project. Although the calculated average staffing is about 2 people, our project has a larger team of 10 members, allowing tasks to be divided and carried out in parallel. Therefore, the actual completion time may differ from the COCOMO estimate depending on task distribution, coordination, and changes in project requirements.

Final Submission Checklist

☒ LOC/KLOC ☒ Mode + reason ☒ Effort ☒ Duration ☒ Staffing ☒ Cost ☒ Interpretation